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Article *in* IEEE Transactions on Instrumentation and Measurement · January 2023

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State-of-the-Art in Data Gloves: A Review of Hardware, Algorithms, and Applications

Mingzhang Pan, Yingzhe Tang, Hongqi Li*, *Member*, *IEEE*

***Abstract*—Data glove is one of the devices that can express human intention via detecting hand motion with high-precision. In recent years, flourished applications have been driven by data gloves from human-machine interaction (HMI) to biomedical sciences. To understand the latest research status and grasp the development direction, this study provides a systematic overview of the state-of-the-art for data gloves. Specifically, we take the research topics of hardware design, algorithms for gesture recognition, and the application areas as the entry point. Detailed elaboration of key algorithms techniques, from data processing to classification, for hand gesture recognition is made and quantitative analysis of the recent research achievements is summarized. We found that hand motion detection with data gloves is mostly employed for gesture recognition, and the majority of studies are focused on the performance validation of the gesture recognition, in particular for static gestures. A systematic illustration of the current research challenges as well as the future directions of related data gloves are finally discussed.**

***Index Terms*—Biomechatronic, human-machine interaction (HMI), data glove, hand gesture recognition, sensors.**

I. Introduction

HUMAN machine interaction (HMI) study explores the cooperative and coupling mechanism between the users and employed devices. It has always been enthusiastic for researchers, in particular with the recent development and introduction of artificial intelligence (AI) technology. Signals related to the physiology, appearance, and even behaviors of a user can be recorded for human intention interpretation in HMI field [1]. As one of the most direct interaction forms in people's daily life, hand gestures (HGs) are attracting increasing attentions.

The data analysis of HGs is used in many engineering applications, ranging from the gesture recognition in HMI to the biomedical sciences. Currently, the technology of gesture recognition is receiving the most popularity in virtual reality among the HMI fields [2]. In addition, how to transform gestures into text is also being a research hotspot [3], [4]. In biomedical sciences, patients are required to perform gesture tasks, whereas physicians assess their hand function ability through the collected data and offer the targeted recovery plans with the aid of a computer [5], [6].

To date, the acquisition and classification of HG data is still challenging and remains a research priority. In general, data acquisition of HG is implemented in two kinds ways named of vision-based and wearable device-based. Vision-based systems capture images or videos as inputs through a camera, with recognition accuracy ranging from 69% to 98% [7]. However, the recognition accuracy are sensitive to the environmental factors (e.g., the backgrounds, illumination, invariance, and ethnic groups) [8]. This leads to findings based on a restricted environment that may not be convincing in real environments. In addition, for some experiments, such as hand rehabilitations, vision-based methods may restrict the daily activities of patients due to the camera limitation [9]. Instead, the wearable devices generally calculate the position, orientation, and location of hands with a faster reaction speed. Compared with the former vision-based devices, the wearable ones rely only on intrinsic sensor readings, which offers better robustness to the environment and has achieved about 90% accuracy averagely in most condition [10-13].

Wearable devices have different forms as electromyographic (EMG)-based platforms [11], [14-16], wristbands [12], [17], etc. Due to the excellent accuracy and intuitive processing characteristics of hand movement data, data gloves have received a great deal of attention. These systems can be defined as the instrumented gloves equipped with various sensors that detect hand motions or changes in finger bending, sometimes including the force feedback information, and transmit the data to other electronic devices (usually being computer processors) [1], [5], [10], [18]. As shown in Fig. 1, according to the Web of Science (WOS), a great number of papers related to data gloves have been published each year and an overall steady upward trend of the paper number is demonstrated from 2015 to now.

Despite hundreds of papers have been published, there is few comprehensive literature reviews focused on data gloves. Dipietro *et al.* [19] proposed a detailed survey about glove-based system advances from mid-1990s to 2008 in the medical and computers areas, they first summarized the evolution history of the data glove, then a review of structures with key characteristics of gloves employed in different applications and related technologies were given. However, a further detailed discussion of hardware designs as well as algorithms of glove-based system were not involved. As time

Manuscript received September 12, 2022. This work was supported in part by the Basic Research Programs of Taicang under Grant TC2021JC27, in part by the Guangdong Basic and Applied Basic Research Foundation Under Grant 2022A1515110252, in part by the Fundamental Research Funds for the Central Universities under Grant D5000210969. (Corresponding author: Hongqi Li, e-mail: lihongqi@nwpu.edu.cn).

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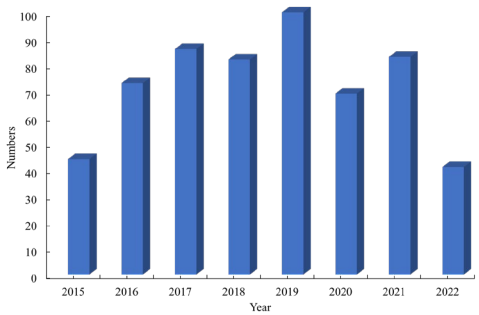

Fig. 1 . Papers published each year from 2015.

goes by, it is not possible to keep up with the latest developments through this study. Nimisha *et al.* gave a literature review of sign language recognition technical in [20], in which the detailed steps of data glove-based gesture recognition and related algorithms were presented. Nevertheless, due to the time limitation and focused point, deep illustration was not revealed. Although Guo *et al.* [21] reviewed kinds of hand gesture recognition methods and compared glove-based recognition with other technologies, a detailed discussion of glove-based approach was not included. In [22], Wang *et al.* made a review which focused on the theme of providing kinesthetic stimuli (i.e., force feedback). The development of haptic gloves was concentrated more on the hardware technology while lacking of the relevant algorithms and software development.

Although the above-mentioned researchers at different stages have conducted several comprehensive reviews, various study priorities and emerging advanced achievement of relevant research still raise an urgent need for a state-of-the-art systematic and quantitative overview of the data gloves. Therefore, aiming to provide the development status, contributors, and trends for the next step of research for those interested in this area, in this paper, we concentrated on conducting a systematic review of data gloves based on the latest representative publications. We conducted literature searches in major reputed publishers, including IEEE, Elsevier, ACM, Springer, etc. We set the search criteria of written in English and published from 2015 to now, with more than 100 representative papers involved.

The main contribution of this work lies in the following three folds:

1) To the best of our knowledge, this study is the first up-to-date systematic review, which works on the relevant technologies of the data glove thoroughly, including the hardware designs, processing algorithms, applications, and future directions of related studies.

2) This work interprets the objective quantitative analysis of the data gloves for structural design, current research topics, and an overview of involved technologies, which differs from the conventional qualitative review.

3) It provides some feedback insights to the hardware designers on how to expand and modify the whole system design based on the usage and existing algorithms.

The remainder of this paper is organized as follows. Section II presents an overview of a data glove system. In Section III, we discuss relevant research on data gloves through the clue of gesture recognition, while Section IV makes a quantitative analysis on the study of data glove area. Section V presents a detailed discussion on the current challenges and future work based on the derived results, with a conclusion provided in Section VI.

II. Research Topics of the Data Glove System

As one of the wearable HG data detection devices, data glove has been studied from different directions.

Fig. 2 shows the basic architecture of the data glove-based systems. In general, the human user wears a glove and performs various static and dynamic gestures by correlating poses and actions, while the data glove captures the data consisting of position, motion, and trajectories of fingers and hand using the sensors located on the gloves. Afterwards, such data is transformed to data set with data processing. Based on the data set, the algorithms model is constructed and employed in different applications for training and online using (gesture recognition, hand assessment, machine control, etc.).

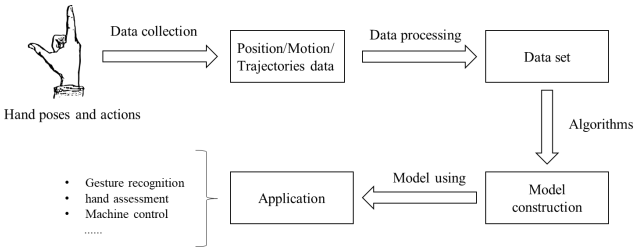

Fig. 2 . Basic architecture of the data glove-based systems.

Based on the architectures mentioned above, the relevant research topics of data glove can be divided into the hardware design (including the materials and structural designs), applicability in different areas, and corresponding methods and algorithms innovation. Specifically, as shown in Fig. 3, for the collected representative research, there is a proportion of 33.67% in studies which concentrated on the applicability of the data glove in the specific areas. The fact that more than one-third of the data share illustrates the current preference of researchers for the breakthroughs of the application domain. Meanwhile, 31.63% of the studies focus on algorithms for the performance improvement and most of them involve the optimization of related classification methods. The ratio is closed to one-third and shows the importance that researchers currently place on the algorithms research. For the remaining studies, 19.39% are related to new materials research and 15.31% are about structural design, respectively. These two

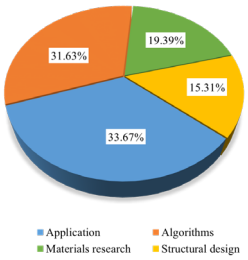

Fig. 3 . The research proportion of data glove focused.

TABLE I
SOME TYPICAL DATA GLOVES

Year	Reference	Type of sensor	Attachment method
2006	Tognetti *et al.* [23]	Conductive elastomer composites	Ink printing
2016	Shen *et al.* [24]	Electrical semi-conducting	Fabric padding
2016	Shen *et al.* [24]	Electrical semi-conducting	Hook and loop
2017	Lin *et al.* [5]	IMU	Elastic tape
2018	Li *et al.* [25]	Bending sensors	Silicone membrane
2019	Glauser *et al.* [26]	Silicone sensor arrays	Textile glue
2019	Huang *et al.* [27]	RGO-coated fibers	Sewing fabric padding
2020	Bhuyan *et al.* [28]	Induction coils	Electrical wires
2020	Nassour *et al.* [29]	Fluidic soft sensors	Silicone adhesive
2020	Mizutani *et al.* [30]	Si-MEMS piezoresistive strain	Adhesion and thermocompression
2021	Jha *et al.* [31]	FBG sensors	Fabric padding
2021	Ayodele *et al.* [32]	Conductive yarns	Weft knit

components together account for 34.70%, which can be referred to as efforts in the hardware. In summary, current data glove research are focusing on the hardware design, algorithms, and specialized applications in about similar proportions (i.e., approximately one-third each). As a result, the remainder of this section will be divided into these following three parts.

A. Hardware Design of Data Glove

As shown in Fig. 4, a data glove system always consists of three components: a sensors array, electronics (used for data acquisition, processing, and power supply), and a support for the sensors based on various attachment techniques [19]. The sensors array is able to transform the hand information into electrical signals and then transmit them to the data processing module [21]. The accuracy of sensors array is a key consideration for the data glove design, while different attachment methods between sensors and sensor support affect the comfort and structural complexity of the data glove. Table I lists a summary of different data gloves of [5], [23-32], with a highlight of the sensors type and attachment methods.

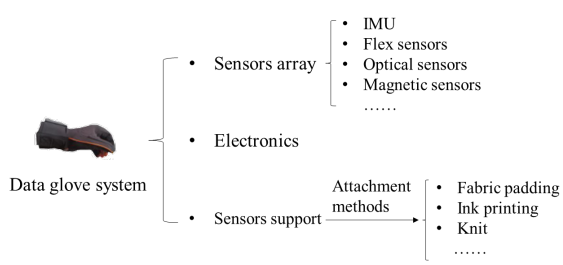

Fig. 4 . Data glove system composition.

Sensor types used in data glove mainly include inertial or inertial measurement units (IMUs) [5], [33-38], flex sensors (e.g., bending ones) [25], [39-41], magnetic sensors such as induction coils [28], and optical sensors [31], [42-46], etc. IMUs that mainly consisting of gyroscopes and accelerometers, are commonly used to measure the acceleration, inclination, and vibration of an object because of their intuitive motion detection characteristics and low cost. Some data gloves in research are composed of exclusively of IMUs [5], [33-35]. Although IMUs are convenient to use, they suffer from a known problem that the rotation angles are computed by integrating inertial signals, which leads to the accumulation of error over time [31]. Recently, a combination of IMUs and other sensors is becoming a new trend. Li *et al*. [36] combined IMUs with optical curvature sensors and soft hall-effect force sensors to design a glove system for a robotic hand teleoperation, and the functionality of the proposed system has been proven in a grasp teleoperation test with very little training for users. Fang *et al*. [37] built a multimodal sign language dataset with inertial-based data gloves, Kinect, and microphone for dynamic gesture recognition. The obtained multimodal data achieve a better result than that based on the single modal. Considering the magnetic field factor, Li *et al*. [38] proposed a multi-sensor data fusion algorithm, for which the state switching extended Kalman filter (EKF) was employed to fuse the inertial sensor data to estimate the accurate finger movement posture. Nevertheless, such fusion still has a deficiency that lacks the hand displacement information.

Besides, flex sensors are attracting the research interest since they are light, compact, low-cost, and more robust. Many studies have captured hand motion with flex sensors and arm positions with IMUs at the same time [25], [39-41]. However, the sensitivity of these sensors is still low for small angles. Plus, the repeatability is rather poor, and the resistance is quite susceptible to changes in temperature [47]. In [28], Bhuyan *et al.* used the induction coils as sensors to measure local hand motions through the related finger bending with the induced voltage measurements based on electromagnetic induction, the cost was brought down by compromising precision to some extent. Alternatively, optical sensors are widely used because of their good accuracy and precision. Islam *et al.* [42]chose an infrared ray (IR) sensor fixed on data glove to detect hand motion, the design is good for maintenance, which is essential for the further promotion. In [43], Li *et al.* designed a data glove with a new type of intensity-modulated optical fiber sensor, which obtains a low error when capturing the hand posture with a low price. Fiber Bragg grating (FBG) sensors constructed using photosensitive optical fibers have also been concentrated on in [31], [44-46] due to their lightweight

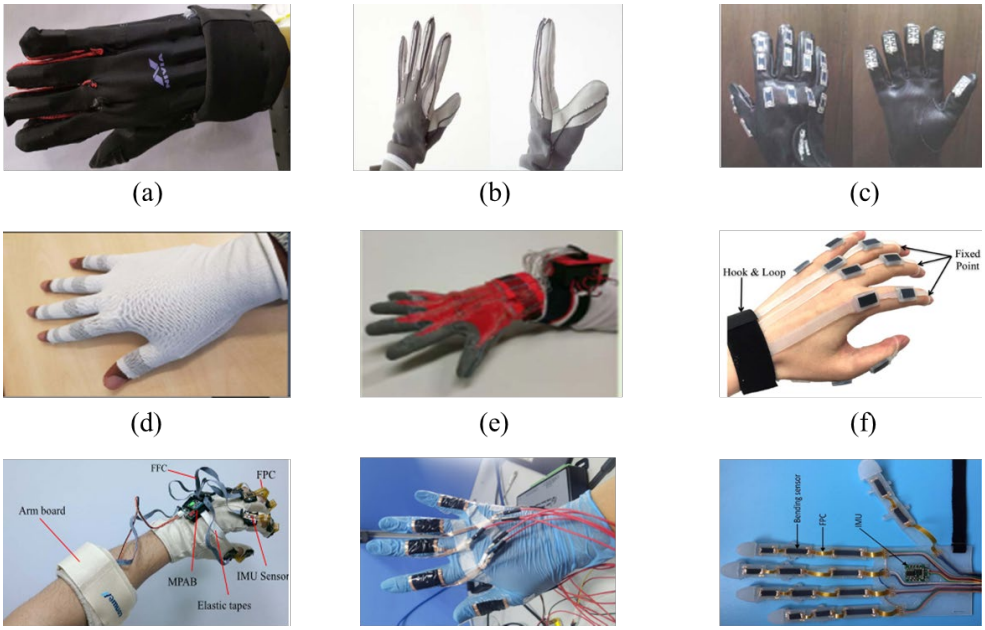

Fig. 5 . Several typical data gloves with relevant attachment methods. (a) From [31]. (b) From [55]. (c) From [60]. (d) From [32]. (e) From [29]. (f) From [24]. (g) From [5]. (h) From [27]. (i) From [25].

characteristics, extremely sensitive to strain, and immune to electromagnetic interference (EMI). Moreover, no hysteresis of capacitive sensor makes them a choice for data glove design recently [26], [48], [49].

Material is an important factor considered in the performance of the sensors. With the development of materials science, many materials research for sensors have emerged. The carbon-based fibers are most commonly used as the sensor materials. In [50], a highly stretchable strain sensor was proposed using a graphene nanoplatelet (GNP) layer and a CB/single-walled carbon nanotube (SWCNT) to synergetic form the conductive network, which showed a lower detection limit and fine linearity. Based on it, Zhang *et al.* [51] proposed a data glove and a data band, which was able to measure the joints movements of shoulders joints, elbows joints, wrists joints, metacarpal and proximal joints of fingers. The proposed system showed a superior advantage of wearability and correlation of human dynamic motions characteristics. In [49], Pan *et al.* proposed a low-cost glove based on a capacitive sensor array, which used a durable silver-MWCNT composite material to achieve both high conductivity and great mechanical robustness. A CDMA-like period modulation multi-capacitive-sensor interface circuit was further used to achieve the low power [52]. The pyrolytic graphite sheets (PGS) sheets are proven to be flexible, synthetically produced, uniform, highly oriented, and inexpensive [53]. Accordingly, Kanokoda *et al.* [54] designed a glove to achieve gesture prediction with a time delay neural network (TDNN) model. Results showed a perfect gesture prediction with no obvious lag. The carbon nanotubes (CNT) are gaining popularity due to their high flexibility, sensitivity and medical compatibility. To better transfer the produced stress, the CNT-based sensors are always mixed by polymers instead of purely composed from CNT. Corresponding demonstration has been proposed such as the soft-thermoplastic polyurethane (S-TPU) and graphene-based electrically conductive thermoplastic polyurethane (C-TPU) in [55], the styrene-butadiene-styrene rubber (SBS) and the C-TPU in [56]. Reduced graphene oxide (RGO) is also attractive to researchers. In [27], Lv *et al.* proposed a reduced graphene oxide (RGO) strain sensor based with encapsulation method, which owns a good linearity, reproducibility and fast response. Huang *et al.* [57] prepared a type of RGO-coated fiber sensor and sewed it onto a textile glove, which realized both a large strain range and a high gauge factor (GF) by encapsulation. Moreover, fluidic soft sensors have been widely used in wearable devices for human motion capturing. In [58], the silicon sensor using an electrolyte solution of potassium iodide (KI) mixed with glycerol (Gly), and the proposed KI-Gly solution showed high linear proprieties and stability of sensor sensitivity. Based on this, Nassour *et al.* [29] proposed a versatile soft sensing glove innovatively, results presented great robustness to the environment and shock. Ionic polymer metal composite (IPMC) is a type of electro active polymer (EAP) with per fluorinated polymer membrane of Nafion or Flemion, and has been pursued for its actuation capabilities. Specifically, Bhattacharya *et al*. [59] explored the sensing capabilities of (IPMC) as a bend sensor. Though the output generated seemed nonlinear, the use of some polymers led to the significantly improved and more reliable response. In addition, to achieve a high stretchability of sensors, polyurethane (PU) is always chosen as the sensor substrate [27], [30], [50] due to its super elastic, and Shen *et al.* [24] chose the ethylene propylene rubber (EPR) to be the sensor material, which has an excellent sensitivity as well as the repeatability.

As for the attachment methods of sensors, Fig. 5 shows several typical data gloves with kinds of relevant methods.

Fabric padding is the most common attachment method in data gloves [31] (seen in Fig. 5(a)), [43], [54], [55] (seen in Fig. 5(b)). It involves placing strain sensors, such as fiber-optic and flex sensors in between multiple layers of fabric in a textile glove [32]. This design makes the data glove to be a little bulky and might impede some implementation of the action. Ink printing is a way that prints conductive ink on glove at phalanx joint locations. Despite its more lightweight alternative characteristic, conductive inks are vulnerable to environmental degradation, which may cause accuracy problem in some environment [23], [60] (seen in Fig. 5(c)). Knit sensors are another unique way whose support structure can be created in a single knitting process, which might have a great potential in the future [9], [32] (seen in Fig. 5(d)), [61], [62]. In addition, some other methods are used depend on the structure of glove, such as adhesion [29] (seen in Fig. 5(e)), [30], hook and loop [24] (seen in Fig. 5(f)), tape [5] (seen in Fig. 5(g)), [27] (seen in Fig. 5(h)), [59], stretchable membrane [25] (seen in Fig. 5(i)), etc.

Furthermore, some other aspects are also attractive and valuable during the hardware design for researchers. For instance, the price is an important reference indicator for promotion, and some studies have made attempts to the balance of price and practicality in [28], [35], [63], [64]. The environmental factor is also taken into the consideration during the design. For example, underwater operations have high requirements for equipment, to achieve extremely operation, Duan *et al.* [65] constructed a glove system with waterproof, machine learning-assisted SPMP fiber. The fiber consisted of four layers (i.e., the spandex core, the poly vinyl alcohol (PVA) layer, the MXene layer, and the PDMS layer) and adopted an in-situ encapsulation technique, which was able to achieve posture signals acquisition with high fidelity in watery conditions. Antillon *et al.* [10] proposed a smart data glove with waterproofed dielectric elastomer (DE) strain sensors and achieved diver communication, while Xu *et al.* [66] designed an underwater humanoid flexible manipulator motion control system based on a data glove instead of a moving rocker for underwater rigid manipulator operation.

B. Algorithms in Data Glove

To get a stable linear output, data gloves always require a calibration step before use since wearers have different hand size. The specific sensor characteristics and the large variations in shape of different hands are usually also taken into account. Calibration is often equivalent to find model parameters, such as gain, offset, or adjusting cross-coupling effects of a custom hand deformation model [26]. The standard calibration technique is quite tedious, time consuming, and impractical in the case of severely disabled hands. In recent years, many optimization methods have been proposed. In [67], Zhou *et al.* calibrated the glove via artificial neural networks (ANN), which trained with the hand-segment sizes as the input and offset, with a requirement for each sensor as the output. In this way, the employed device is always a 2-D digital camera, which can be easily used by even patients with severe upper extremity disability. However, there are still some limitations existed, for example, no cross-coupling factor is considered. Steffen *et al.* [68] proposed a novel data glove mapping technique to solve the cross-coupling effect, which can achieve a robust calibration by simply moving the worn glove to extreme positions allowed. Wang *et al.* [69] designed a gaussian process regression (GPR) model to infer the calibration function to handle sensor coupling problem, which can achieve both forward kinematics (FK) and inverse kinematics (IK) accuracy. Connolly *et al.* [70] constructed a set of neural network (NN) calibration lookup tables that bypass the requirement for complex calibration. Model proposed increases overall movement accuracy for sensors that have a trait of nonlinearity, such as with bend sensors. Moreover, according to [26], Min-max pose calibration [71], ground truth calibration [72], and IK calibration [73] are among the most common approaches.

After calibration, data acquisition can be carried out. It is necessary to do data processing after data acquisition to get purer data, which usually includes filtering and noise reduction. Using methods such as principal component analysis (PCA) to achieve dimensionality reduction can make data analyze easier. Feature extraction is always performed when performing classification tasks, such as gesture recognition, and a classify model is usually built by some machine learning (ML) algorithms such as support vector machine (SVM), K-nearest neighbor (KNN) or deep learning (DL).

C. Applications of Data Glove

Based on the hand motion data captured by data gloves, different application areas have flourished in recent years. Table II depicts the representative application areas.

TABLE II

DATA GLOVE APPLICATIONS

Application		References
Virtual reality	Virtual environment interaction	[74], [75]
	VR gaming	[76]
	VR surgical training	[77]
Hand function assessment	Assessment	[5], [6], [13], [34], [78], [79], [80], [81], [82]
	Rehabilitation	[83], [84]
Gesture recognition	Glove performance validation	[27], [29], [57]
	Machinery control	[85], [86], [87], [88]
	Sign language recognition	[10], [89], [90], [91], [92] [93], [94], [95]

As shown in Table II, in virtual reality (VR) and augmented reality (AR) applications, tracking of hand motion allows for good interaction with the virtual environment [74], [75] and other tasks, like VR gaming control [76] and VR Surgical Training [77], etc. With the fast and highly accurate monitoring of hand movements, data glove becomes a good choice for hand function assessment as well, which is essential to diseases like paralysis or poststroke and become a focus of research. Many studies have concentrated on hand health assessment [5], [6],

[13], [34], [78-82], and complementary rehabilitation [83], [84] based on data glove.

Indeed, in all related application areas, glove-based gesture recognition has been the mainstream. Gesture recognition always used in experiments for the accuracy validation of the glove, such as [27], [29], [57]. With the development of artificial intelligence, some researches of machinery control are proposed, where Müezzinoğlu *et al.* [85] attempted to control unmanned aerial vehicles with hand gesture. Simão *et al.* [86] and Zhang *et al.* [87] controlled the robot arm with hand gesture based on data glove to achieve some basic grasp action. Shankar *et al.* [88] made some experience in using gestures to control home appliances based on data glove to help the life of disabled person. In addition, sign language recognition is an important direction of hand gesture recognition, which brings convenience to the communication of deaf-mute. In this field, Ahmed *et al.* [89] proposed a Malaysian Sign Language (MSL) recognition framework based on data gloves. Lee *et al.* [90] did research focused on repetitive gesture detection, which has practical significance for sign language conversation in the real world. Moreover, many researchers attempted to combine data gloves with smartphones to convert sign language into text or speech [91-95]. At the same time, data glove-based gesture recognition provides an option for sign language communication in harsh environments like conditions of underwater [10].

III. Key Technique of Algorithms for the Glove-based Hand Gesture Recognition

Hand gestures can be divided into static and dynamic hand gestures. Static hand gestures, referring to some particular hand configurations and poses without hand movement. Dynamic hand gestures are moving gestures that include movements of fingers or hands even arms, which include two kinds of gesture types (i.e., isolated gestures and continuous gestures). For both the static and dynamic gestures, the final goal of the gesture analysis is to extract characteristic features describing the movement of hands and fingers. It seems reasonable to use data that are as directly related to features describing gestures as possible to test the feasibility of the task of recognizing the hand gestures of the human body language [1]. Data gloves are widely accepted, as the data recorded are directly related to the hand movements.

The basic steps of data glove-based gesture recognition are summarized in Fig. 6. Gesture data is first acquired by the data glove and then data preprocessing and feature extraction are used to realize the functions, such as filtering and dimensionality reduction. Finally, the adopted classifier is trained with the processed data to obtain a classification model to achieve the classification and recognition of gestures. Since the previous section II has shown the important roles of gesture recognition in data glove application, here we concentrate more on relevant technological developments.

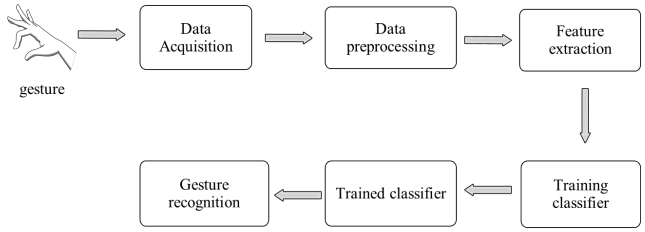

Fig. 6 . Basic step of hand gesture recognition.

A. Data Acquisition

As shown in the Fig. 7 except for the thumb finger, each finger has three joints, i.e., a proximal interphalangeal (PIP) joint, a metacarpophalangeal (MCP) joint, and a distal interphalangeal (DIP) joint, respectively. The thumb has only two joints, i.e., the interphalangeal (IP) joint and metacarpophalangeal (MP) joint, where the IP joint can be approximately treated as the PIP joint of other fingers. Note that the MCP joint has two degrees of freedom (DOFs) for bending/extension and abduction/adduction, while the PIP and DIP joints only have bending/extension DOF [57]. Therefore, the hand shape can be mainly determined by the motion of totally 14 finger joints in one hand, and the sensors are usually used to measure the joint angles. In addition, as the motion of the DIP joint is driven by the PIP joint and cannot be rotated independently [96], the bending angle of such joint can be inferred through that of the PIP joint, many studies focus on the MCP and PIP measurements instead of three finger joints [57], [97]. It should be noted that the majority of data gloves only focus on finger flexion-extension while ignoring abduction-adduction movements because such the movements' measurement requires sensors placed between fingers, which may cause discomfort and impact the degree of hand freedom. Nassour *et al.* proposed a kind of soft sensor to solve this challenge in [29].

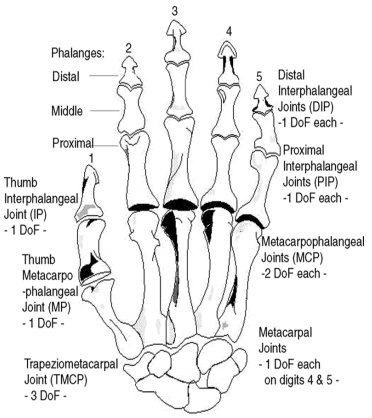

Fig. 7 . Schematic diagram of the finger joints [19].

The data of hand finger joints are commonly collected by IMUs or flex sensors. IMUs are placed on both sides of the joint, and an algorithm is employed to calculate the flexion angle. A flex sensor is usually embedded in the data glove and then the change of its resistance due to finger bending can be converted into angle-related values. Sometimes hand gestures involve arm movement, and IMUs need to be strapped on the arm to obtain more detailed hand motion data.

B. Data Preprocessing

The main objective of data preprocessing is to obtain a representative data set. Filtering the raw data can remove redundant information from the data, whereas the further

determining or even enhancing the important information is also essential. Thus, the denoising of acquired signals is usually inevitable especially when the accelerometer data was used [98]. In the research of Dutta *et al.* [78], raw sensor output was denoised using a savitzky-golay (SGolay) filter to maximize the precision of data without affecting the trend of the signal, and a 5-point moving average (MA) filter was also used to smooth the data.

For the data itself, those missing values also need to be processed with the option to delete or replace them with other values according to different situations. Scaling is required when the ranges of characteristics are different, and common methods include normalization and standardization. Although the greater the amount of data leads to the better persuasion of classification, too much quantity comes at the cost of a long time. Therefore, a trade-off between the amount of data and time-cost is needed. Pławiak *et al.* [1] applied PCA to test if the reduction of data may increase classification sensitivity. In the experiment of Simão *et al.* [86], a classification accuracy with 25% of gesture data is higher than that with the whole gesture data. Calado *et al.* [39] concentrated on obtaining a good accuracy with minimum number of sensors. Chan *et al.* [99] proposed a method that can check and removing sensors not important enough, thus removing data channels from the input with a decent accuracy for gesture classes.

In regard to time series, it is also essential to find important information from the large amount of data, and data segmentation is an important technique [83], [97], [100].

C. Feature Extraction

Feature extraction is a dimensionality reduction step, where the initial set of data is reduced to more manageable community features while maintaining the accuracy and completeness of the description of the original set of data.

Typical features include time-domain features (such as the mean standard deviation), frequency-domain features, and time-frequency- domain features, respectively. Sometimes features need to be selected based on parametric models. Commonly used methods of feature extraction include PCA [34], and linear discriminant analysis (LDA) [48]. Manual feature extraction based on experience is also currently used. For instance, Chen *et al.* [83] chose five features in their research, including the mean absolute value (MAV), root mean square (RMS), waveform length (WL), variance (VAR), and standard deviation (SD). In addition, DL is gaining popularity due to its ability to automate feature extraction [101].

Feature selection aims to select the optimal feature subset from the original features to improve the accuracy, which mostly includes three approaches: filter methods, wrapper methods, and embedded methods. The specific methods include backward selection [34], ReliefF [78], etc. Additionally, Calado *et al.* [39] chose a filter method based on the correlation between features.

D. Classification and Evaluation

The purpose of classification is to obtain a function or classification model from a given set of manually labeled classification training samples. Then, when the new data are obtained, the predictions can be made.

The current common classification algorithms include naive bayesian (NB) [10], [40], [85], [93], [100], [102], [103], logistic regress (LR) [10], [103], [104], decision tree (DT) [10], [83], [103], [105], SVM [1], [9], [10], [39], [40], [78], [83], [85], [93], [100], [102-104], [106], [107], KNN [1], [9], [39], [40], [63], [83], [85], [102-104], [106], [107], ANN [39], [65], [103], etc. Specifically, Fang *et al.* [35] proposed the extreme learning machine (ELM), an algorithm with a fast learning-speed, independent from implementation, and minimal human intervention. Sarwat *et al.* [13] adopted the extreme gradient boosting (XGBoost) algorithm, which is a gathering strategy with high precision, parallelizable processing and portability. Dong *et al.* [101] proposed a novel dynamic hand gesture recognition algorithm named DGDL-GR which fully utilized a fusion model of convolutional neural network (fCNN) and generic temporal convolutional network (TCN). Moreover, Calado *et al.* [106] proposed a geometric model-based approach for hand gesture recognition, it was proven to have a good interpretability for which DL and ML lacked. In [108], gesture recognition was achieved by a method which employed the least squares method to fit the gesture joint motion data first, and then calculated the corresponding Hausdorff distance (between the gesture data to be recognized and the existing data in the sample library). The gesture recognition problem was thus converted into a similarity curve problem with a high accuracy and robustness. In [109], the gated recurrent unit (GRU) was proposed by Chuang *et al.* to achieve continuous gesture recognition and the results showed a good robustness. Bal *et al.* [110] chose a probabilistic neural network (PNN) to focus on practical usage of dynamic gestures. In [111], a fuzzy logic algorithm was utilized with neural networks, which united their advantages and excluded disadvantages like the ability of fuzzy logic to interpret input. In [112], Simão *et al*. proposed a novel way of solving the issue of classification of out-of-vocabulary gestures within ANNs trained in the generative adversarial network (GAN) framework.

To evaluate models and compare the performance with different methods, unified evaluation criteria are necessary to be determined. The mostly widely used indicators include the accuracy, recall rate, precision, and F1-score. The receiver operating characteristic (ROC) curve-based method and area under the curve (AUC) are also considered in some research. Among all of these, accuracy is the most important metrics. In practical applications, the response time determines the viability of the equipment. However, only a few studies have been concerned with response time in detail [1], [85], [97], [106]. Grützmacher *et al.* [113] proposed a hand gesture recognition system that utilized thread-level parallelism on a many-core DSP architecture to reduce latency. It is worth to notice that sometimes researchers also customize the criteria based on the actual research situation [1].

IV. Current Quantitative Analysis of the Data Gloves Systems

In this section we present a summary analysis of current experimental studies related to data gloves, covering the selection of relevant indicators for the experiments, such as the gestures, subjects, etc., in addition to a categorization analysis of classification algorithms.

As shown in Fig. 8, in the collected studies, the static gesture recognition is still being a mainstream. There is 51.11% of studies aimed at static gestures, while dynamic gestures only account for 31.11%, and the rest 17.78% representative studies included both static gestures and dynamic gestures. In other words, there is a percentage about 70% of studies which related to the static gestures.

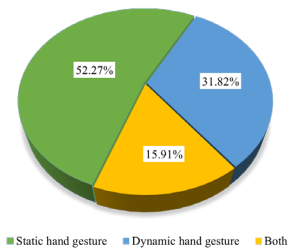

Fig. 8 . Hand gesture recognition in studies focused.

No matter static gestures or dynamic gestures, the validation of the methods is necessary and it is usually related to several factors: the number of gestures, the number of subjects, and the total sample size.

As shown in Fig. 9, the smallest number of validated gestures is 3 in [54], where the largest number being 31 in [99]. In most studies, the number of gestures was approximately 10-20 [10], [25], [27], [78], [90], [104]. Moreover, the number of specific static and dynamic gestures types are listed in Fig. 10 and Fig. 11, respectively. Overall, the number of static gestures focused now is larger than that of the dynamic ones. We see that from the Fig. 10 that the focused number exceed 15 in studies occupies nearly half, which is more than that of dynamic gestures in Fig. 11. In addition, the number is averaged 20 of static gestures in Fig. 10 while the number of dynamic gesture types was below 10 averagely in Fig. 11. The above three figures show that the number of dynamic gesture types in studies are less compared to static gestures. The average number of 20 for static gesture is somewhat representative, while the variety of dynamic gestures may be underrepresented.

As for subjects' numbers, the average number was approximately 20 among all references, as illustrated in Fig. 12. Notably, studies concentrating on the hardware design of data glove [29], [35] or justification of new approaches always have fewer subjects while those related to clinical investigations [5], [80], [82] prefer to adopting more examinees.

Table III summarizes the sample number adopted in the latest research and Fig. 13 shows the specific percentage. Studies whose number of samples less than 1000 are approximately occupied 21.05%, and 26.32% of studies have more than 10000 samples. The sample size of most studies is ranged from 1000 to 10000, with a most percentage of 52.63%. In addition, total sample size is generally at a low level.

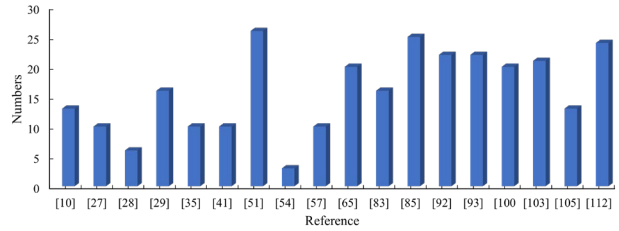

Fig. 10 . The number of static gestures in several studies focused.

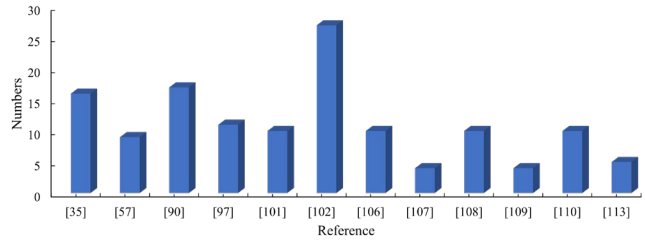

Fig. 11 . The number of dynamic gestures in several studies focused.

The study of classification algorithms plays an important role in the field of data glove, especially in glove-based gesture recognition. The related evaluation conditions have been summarized in Fig. 14. As shown in Fig. 14, most research chose 3 to 5 algorithms to make a comparison. Generally, studies concerned more with hardware design used fewer algorithms [9], [13], [29], [65], while those focused on the algorithm research employed more for comparative verification [101], [103], [106]. Alrubayi *et al.* chose the highest number of algorithms in [103], 10 algorithms including KNN, ANN, SVM, DT, LDA, LR, random forest (RF), NB, gradient boosting (GB), and stochastic gradient descent (SGD) were selected to evaluate the pattern recognition model for MSL gesture recognition. Table IV lists the best classification algorithm adopted by the relevant studies and their optimal accuracy. As we can see that the majority of studies are with 90.00% accuracy above. Among all the algorithms, classical recognition algorithms such as SVM, KNN are still effective in the data glove areas, and always served as a control benchmark. The most popular algorithms are SVM for its excellent generalization and simple principle, as shown in Fig. 15.

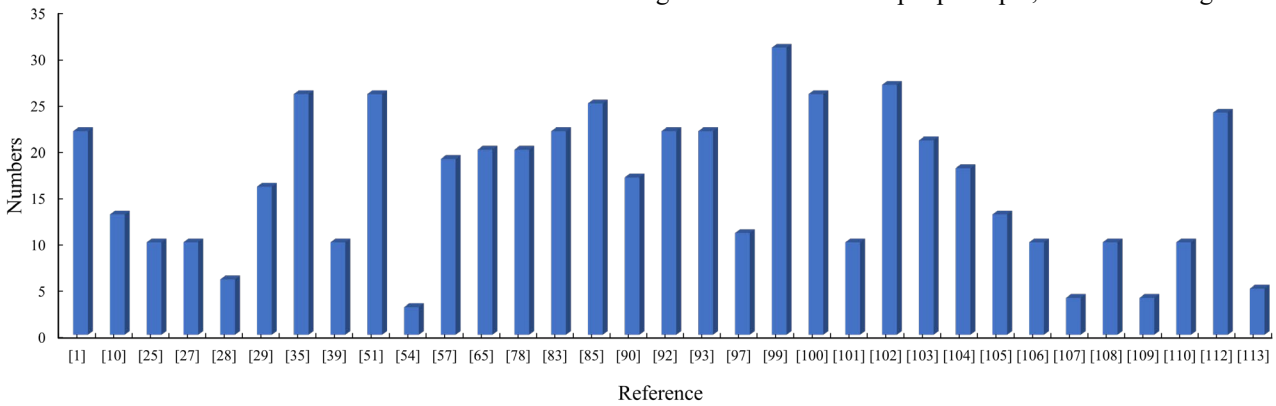

Fig. 9 . The number of gestures in several studies focused.

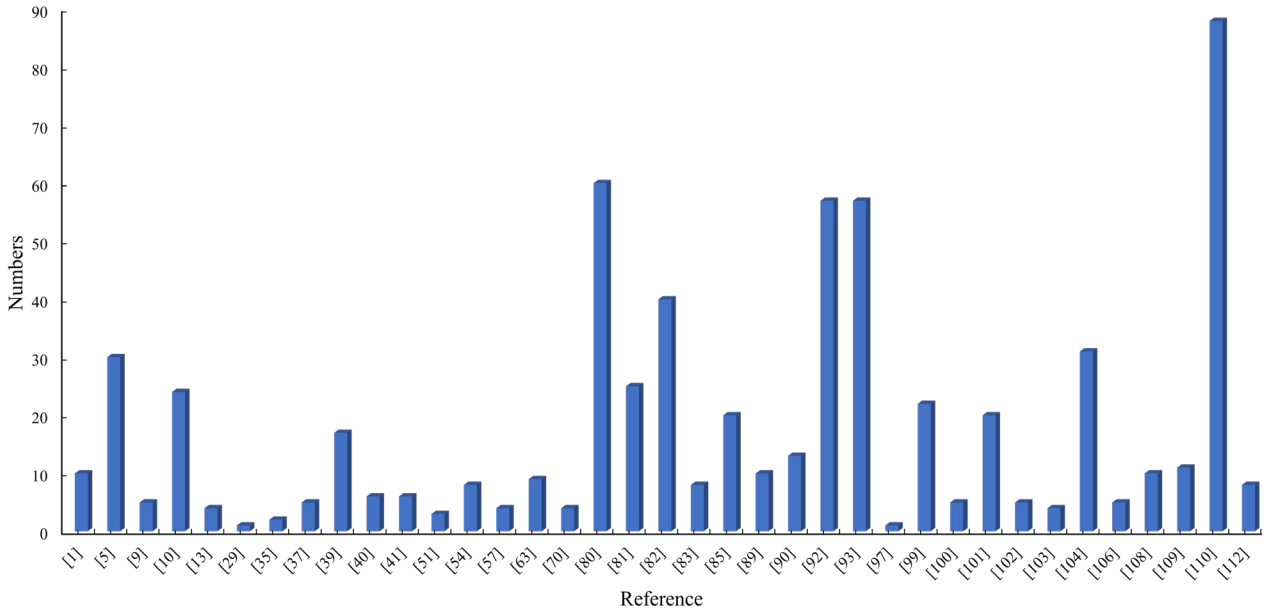

Fig. 12 . The number of subjects in several studies focused.

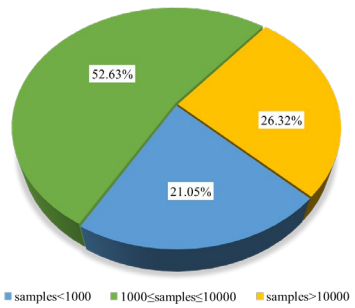

Fig. 13 . The sample number of research.

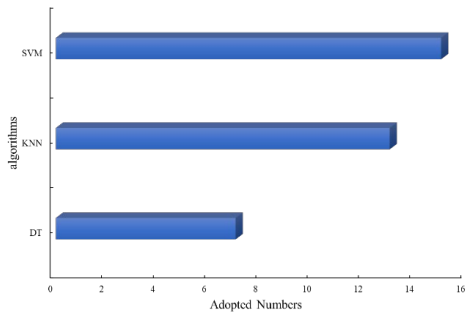

Fig. 15. Commonly employed algorithms.

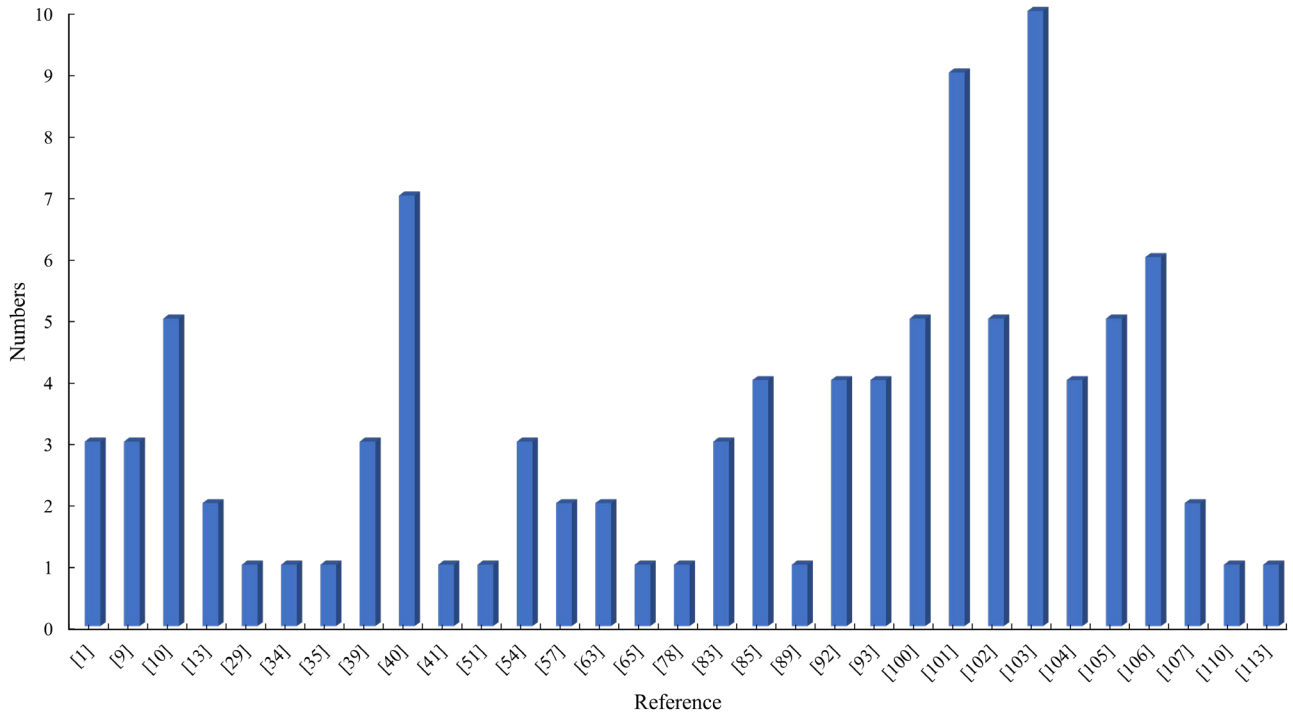

Fig. 14 . The number of algorithms used in several research focused.

TABLE III
THE SAMPLE NUMBER OF SEVERAL RESEARCH

Year	Reference	Samples numbers
2016	Plawiak *et al.* [1]	2200
2018	Mummadi *et al.* [93]	1250000
2019	Huang *et al.* [57]	2160
2019	Chuang *et al.* [109]	4488
2019	Simão *et al.* [112]	2400
2020	Lee *et al.* [97]	880
2020	Yuan *et al.* [40]	39630
2020	Fu *et al.* [41]	300
2020	Zhang *et al.* [51]	1000
2021	Calado *et al.* [39]	10596
2021	Dong *et al.* [101]	1127
2021	Pezzuoli *et al.* [102]	810
2021	Müezzinoğlu *et al.* [85]	49000
2021	Alrubayi *et al.* [103]	2100
2022	Lv *et al.* [27]	315
2022	Duan *et al.* [65]	3000
2022	Antillon *et al.* [10]	15080
2022	Lee *et al.* [90]	1072
2022	Simoes Dias *et al.* [100]	1950

V. DISCUSSION

The research and development of data glove have received a great attention for their convenience and accuracy in gesture detection. Nevertheless, the existing work still have many challenges. First of all, regarding the hardware, many current studies are concentrated on the accuracy improvement through structure design and materials. Although high accuracy is now generally available, the cost factor is still an open issue and challenging in data gloves. Some research has made efforts in this aspect [28], [64], and there is still a long way for the final practical application, especially the cost of manufacturing high-precision sensors. We hope that continued development of materials science can facilitate the solution of such the mentioned problem. In addition, the choice of materials for the manufacture of data gloves should be based on the application environments. For instance, when taking the clinical application [33] into account, the comfort of the glove is a factor which cannot be ignored, specifically it is not permitted to cause any damage to patients' hands. Besides, when the application is working underwater [10], [65], the waterproof nature of the materials needs to be the first consideration. Soft data gloves are able to better fit the hand, increase the user comfort, and help improve accuracy. Thus, soft sensors [24] and knit sensors [32], [61], [62] have been a development direction. Moreover, most data gloves still need to be combined with an external processor to achieve data analysis, which limits its scope of application. Embedded data analysis with data acquisition in data gloves will also be a future focused direction. Such the implement can be achieved with the help of the edge device [48], which is important for on-site decision-making.

TABLE IV
CLASSIFICATION ALGORITHMS OF SEVERAL REPRESENTATIVE STUDIES

Year	Works	Best algorithm	Accuracy
2015	Grützmacher *et al.* [113]	DTW [a]	70.50%
2016	Plawiak *et al.* [1]	SVM	κ = 98.24% [b]
2016	Sekar *et al.* [64]	Template matching	94.50%
2016	Luzhnica *et al.* [104]	LDA+LR	98.20%
2017	Mummadi *et al.* [92]	RF	92.00%
2017	Lv *et al.* [107]	SVM	93.19%
2017	Guo *et al.* [108]	Template matching	98.00%
2018	Fang *et al.* [35]	ELM	Static gesture: 89.59% Dynamic gesture: 82.50%
2019	Kanokoda *et al.* [54]	ANN	90.40%
2019	Huang *et al.* [57]	Static gesture: NN Dynamic gesture: DTW	Static gesture: 98.50% Dynamic gesture: 82.50%
2019	Chuang *et al.* [109]	GRU-based	97.27%
2019	Simão *et al.* [112]	GAN	90.20%
2020	Yuan *et al.* [40]	DFNN [c]	ASL:99.30% CSL:96.10%
2020	Nassour *et al.* [29]	RF	89.50%
2020	Zhang *et al.* [51]	RBFNN [d]	93.33%
2020	Zanzarukiya *et al.* [95]	CNN	97.76%
2021	Ayodele *et al.* [9]	CNN	88.27%
2021	Calado *et al.* [39]	ANN	93.91%
2021	Chen *et al.* [83]	SVM	Finger gesture: 93.32% task-oriented gestures: 89.40%
2021	Müezzinoğlu *et al.* [85]	SVM	98.02%
2021	Dong *et al.* [101]	DGDL-GR	98.69%
2021	Pezzuoli *et al.* [102]	RF	99.70%
2021	Min *et al.* [105]	ResNet [e]	99.20%
2021	Bal *et al.* [110]	PNN	92.70%
2022	Duan *et al.* [65]	MLP [f]	98.10%
2022	Antillon *et al.* [10]	SVM, LR	98.00%
2022	Dutta *et al.* [78]	SVM	92.00%
2022	Lee *et al.* [90]	RNN-based [g]	94.21%
2022	Simões *et al.* [100]	RF	96.15%
2022	Calado *et al.* [106]	Geometric-based	90.00%

[a] Dynamic time warping, [b] κ is is a factor defined by the authors based on the number of erroneous classifications, [c] Deep feature fusion network, [d] Radial basis function neural network, [e] Residual neural networks, [f] Multilayer perceptron, [g] Recurrent neural network.

Second, algorithms influence the interpreting performance and in the currently data glove areas, traditional classification algorithms are still popular because of the small databases. Notably, in recent years, increasing number of researchers have started to incorporate DL [40], [51], [97] to find a breakthrough. Meanwhile, useful feature selection ensures effective implementation of the algorithm model. Different application

scenarios may require different features, while fewer papers have dived into it. Therefore, one of the future works is to explore the features with higher degree of correlation, continuously optimize the performance of the algorithm, considering kinds of factors based on practical. It is worth noting that when it comes to sign language recognition, different national sign languages have their own characteristics, differences such as similarity of gestures, number of dynamic gestures, and amount of data collected may affect the performance of the classifier [100]. For instance, American sign language (ASL) mainly focuses on the palm posture and fingers bending, while has little to do with fore and upper arms, and most ASL symbols don't need sequential representation. Compared with ASL, motion data of Chinese sign language (CSL) are always complex, sequential and disturbed from each other [40]. The impact of different learning rates and batch size on their precision is different. Therefore, this difference needs to be taken into account when designing the algorithm.

Third, for the application, hand gesture recognition is the main research direction of the current data glove. Static gestures are the focus due to their ability to convey enough meaning with simplicity. Current glove-based gesture position determination and tracking are mainly achieved by the IMU through dead reckoning technique [114], and several other approaches with the microelectromechanical systems (MEMS) [115], and magnetic positioning system (MPS) [116]. Little attention has been paid to the actual spatial location of the gestures, whereas the 3D hand position can be actually acquired with ultrasonic receiver array [117]. On the other hand, in order to achieve richer gesture expressions, more attention needs to be devoted to dynamic gestures, especially continuous dynamic gestures, only few studies have made efforts in this area currently [90]. Indeed, within the gesture recognition in data gloves, several mature techniques can be naturally integrated. For instance, radio frequency identification (RFID) is an effective automatic identification technology that enables simple proximity detection of tagged items as well as low-cost information sharing. It has been successfully used in the target object tracking [118], and indoor location [119]. Recently, RFID-based gesture recognition has attracted researchers' attention. Sedighi *et al.* [120] proposed an environment interaction method for the visually impaired. Based on the passive RFID tags, Asadzadeh *et al.* [121] achieved motion pattern tracking, while Cheng *et al.* [122] recognized six gestures by examining and evaluating the phase trend of the backscatter signals.

Furthermore, the sample size of most current papers about hand gestures is still low when it comes to practical application scenarios. Gesture number is an important factor in the relative research. Although the variety of static gestures is rich, the number of dynamic gestures is still in a relatively small range. In addition, the subject number in most studies keeps still low, which may have the robustness problem to the environments when performing practical applications. Moreover, the data acquisition of data glove-based is complex owing to the high equipment price and the need for calibration before operation, which leads to the phenomenon that there is no public dynamic gesture dataset for data glove-based gesture recognition [123], [124]. Obtaining a correct and suitable dataset size for an application and creating adequate databases are not easy and should become an urgent issue to be resolved in the near future. To avoid complex and boring calibration steps as well as open data glove use to those with limited joint mobility, the method of calibration before use is also a direction need to be studied.

Finally, as for the evaluation metrics, accuracy is indeed an important index to test the validity of the classification and almost all data glove-based researches have considered this metric. It can be seen that data glove-based gesture recognition generally has good performance advantages, which demonstrated the high accuracy of the data gloves. However, considering application scenarios such as HMI, other metrics such as response time also need to be taken into account. To date, most of the papers have not taken a deep focus into it.

VI. Conclusion

As a common form of hand gesture recognition, studies based on data glove has been developed rapidly in both human-machine interaction and biomedical sciences. This review focuses on the latest development status and aims to provide most important ideas for those interested in this field. The hardware design, algorithms, and application of data gloves are thoroughly reviewed first. A detail introduction to gesture recognition, which is the core area of data glove is then presented. Through the quantitative analysis, we finally draw out the current research trends and present our views on the future direction of relevant development.

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